So, what is the quantum Fourier transform?

Kai Sandberg, October 2026

First, recall the classical discrete Fourier transform (DFT). Suppose we are given a vector $\vec{c} \in \mathbb{C}^n$. The Fourier basis for $\mathbb{C}^n$ is $\{\boldsymbol{f_k}\}_{k=0}^{n-1}$, where

$$\boldsymbol{f_k} = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 \\ e^{2\pi i k/n} \\ e^{2\pi i k(2)/n} \\ \vdots \\ e^{2\pi i k(n-1)/n} \end{pmatrix}$$

In other words, any vector $\vec{c}$ can be written as $\vec{c} = \sum_{k=0}^{n-1} a_k \boldsymbol{f_k}$ with $a_k \in \mathbb{C}$.

- For example, consider $\mathbb{C}^2$. The Fourier basis is then $\boldsymbol{f}_0 = \frac{1}{\sqrt{2}}(1,1)$ and $\boldsymbol{f}_1 = \frac{1}{\sqrt{2}}(1,-1)$, which clearly is a basis for $\mathbb{C}^2$.

Concisely, 'taking' the DFT of a vector $(c_0, ..., c_{n-1}) \in \mathbb{C}^n$ means to output the vector $(a_0, ..., a_{n-1}) \in \mathbb{C}^n$ with a_i defined in the sum above (the Fourier coefficients). The DFT is often expressed as a unitary matrix, which takes in $\vec{c}$ and spits out $\vec{a}$.

So, what is the QFT?

Recall that any pure quantum state of q qubits, $|\psi\rangle = \sum_{k=0}^{n-1} c_k |k\rangle$ with $n = 2^q$, can be expressed uniquely by $\vec{c} = (c_0, ...c_{n-1}) \in \mathbb{C}^n$. The QFT is simply a matrix U that takes in $|\psi\rangle$ and outputs $(a_0, ..., a_{n-1}) \equiv \sum_{k=0}^{n-1} a_k |k\rangle$.

- What are a_k, explicitly? Since $\vec{c} = \sum_{k=0}^{n-1} a_k \boldsymbol{f_k}$ simply dot by $\boldsymbol{f_k}$ on the right. By orthogonality, $\langle \vec{c}, \boldsymbol{f_k} \rangle = \overline{a_k}$, so $a_k = \boldsymbol{f_k}^\dagger \vec{c} = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} e^{-2\pi i k j/n} c_j$, which is the famous formula you see whenever learning about QFT.

Thus,

$$U_{\text{QFT}}|\psi\rangle = U_{\text{QFT}} \sum_{k=0}^{n-1} c_k |k\rangle = \sum_{k=0}^{n-1} a_k |k\rangle = \sum_{k=0}^{n-1} \Big(\frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} e^{-2\pi i k j/n} c_j \Big) |k\rangle$$

> In summary, the QFT is a matrix that takes in a quantum state (determined by $\vec{c}$), and outputs a new state whose amplitude vector $\vec{a}$ are the Fourier coefficients of $\vec{c}$.

Remark: Most textbooks and papers use an opposite convention - their QFT matrix takes in $\vec{a}$ and spits out $\vec{c}$. In this case, our U_{QFT} is their U_{QFT}^{-1}.

Why is QFT useful?

Since $U_{\text{QFT}} \vec{c} = \vec{a}$ (change of basis), then

$$U_{\text{QFT}} = \begin{bmatrix} | & | & & | \\ \boldsymbol{f_0} & \boldsymbol{f_1} & \cdots & \boldsymbol{f_{n-1}} \\ | & | & & | \end{bmatrix}^\dagger,$$

which is the $n \times n$ DFT matrix! So... the QFT is just the DFT? Yes. Why is it useful? Applying the QFT circuit on a superposition state computes the DFT on each component *simultaneously*. In other words, we can perform the $n \times n \equiv 2^q \times 2^q$ DFT using only q qubits. The caviat (as always), is that we can't read out the amplitudes of the quantum state; they are encoded as probabilities upon measurement.

QFT as a quantum circuit

How might we construct U_{QFT} as quantum gates? Let's examine the 1-qubit case.

Exercise 1: Compute $U_{\text{QFT}}|0\rangle$ and $U_{\text{QFT}}|1\rangle$ using the formula above. Hint: Express $|0\rangle$ and $|1\rangle$ as vectors to obtain c_k, and plug into the formula. In this 1-qubit case, does the result look like a familiar quantum gate?

Notice that for 1 qubit, $U_{\text{QFT}} = H$, the Hadamard gate! Unfortunately for us, this only applies for the $q = 1$ qubit case. In general, applying Hadamard gates on q qubits is not equivalent to U_{QFT}.

Let's inspect the 2-qubit QFT matrix:

$$U_{\text{QFT}} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}$$

To determine what gates we should use, let's see the action on the basis states:

$$|00\rangle \mapsto \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

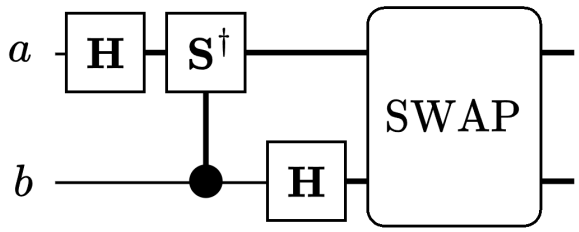
$$|01\rangle \mapsto |-\rangle \otimes \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

$$|10\rangle \mapsto |+\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

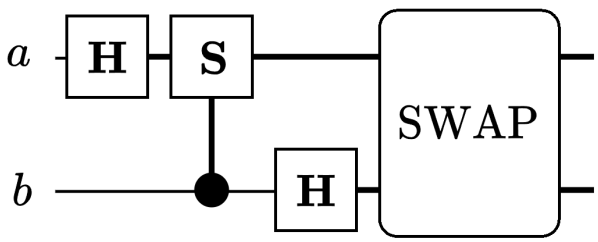
$$|11\rangle \mapsto |-\rangle \otimes \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

Evidently, the first qubit is $|+\rangle$ or $|-\rangle$ depending on the second qubit. The only thing that changes in the second qubit is the coefficient (phase) in front $|1\rangle$. If our initial state was $|ab\rangle$ (where a and b are 0 or 1), then the first qubit becomes $H|b\rangle$, and the second becomes $\frac{|0\rangle + (-1)^a(-i)^b|1\rangle}{\sqrt{2}}$. If we define $S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$ (since S is already a standard quantum gate), the second qubit is $S^\dagger H|a\rangle$, where $S^\dagger$ is applied conditionally IF $b = 1$. Thus, our circuit is as follows:

- First, H then conditional-$S^\dagger$ on first qubit; this gives $(S^\dagger)^b H|a\rangle$.
- Next, H on the second qubit, which gives $H|b\rangle$.
- Finally, use a SWAP gate on both qubits to get the desired order.



Note: Qiskit uses the opposite convention as ours. As such, take the conjugate transpose(†) of the entire circuit. Luckily, U_{QFT}^{-1} satisfies $U_{\text{QFT}}^{-1} = U_{\text{QFT}}^\dagger = U_{\text{QFT}}^\star$, and conjugation ($\star$) doesn't swap gate order. Qiskit's 2-qubit QFT looks like this:



QFT circuit on q qubits.

If we factor out the QFT on q-qubits, or$|a_1 a_2 ... a_q\rangle$, as $|\psi_1\rangle \otimes |\psi_2\rangle \otimes ... \otimes |\psi_q\rangle$, the first factored state is always $|+\rangle$ or $|-\rangle$ depending on a_q. For all subsequent states, the all have the form $\frac{|0\rangle + (\text{phase})|1\rangle}{\sqrt{2}}$. It turns out the $|1\rangle$ phase for qubit a_i turns out to be

$$(-1)^{a_i}(-i)^{a_{i+1}} e^{-(\pi i a_{i+2})/4} ... e^{-(\pi i a_q)/2^{q-i}}.$$

If we let $R_m = \begin{pmatrix} 1 & 0 \\ 0 & e^{-2\pi i/2^m} \end{pmatrix}$, then the gates are as follows:

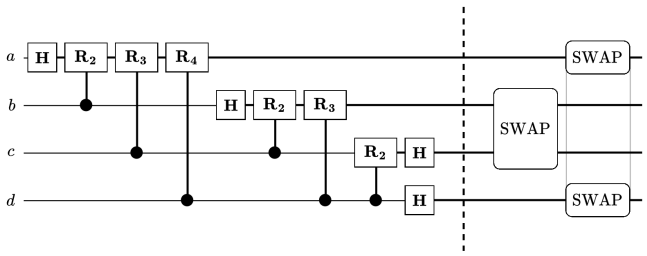
Starting on qubit $j = 1$ up to $j = q$:

- Apply H
- For each $k = j + 1, ..., q$, apply $R_{(k+1)-j}$ controlled by qubit k

Finally, SWAP the order of all the qubits.

If we are not using the QFT state further in the circuit, we don't have to SWAP all the qubits and can just interpret them in reverse order.

4-qubit example:



Again, to match Qiskit's convention, replace each R_i with $R_i^\dagger$.